To thoroughly investigate the phenomenon of return decays, we examine two trend-following trading strategies—TSM and RSM—based on a look-back period of $j = 12$. Figure 3 illustrates the decay in trend-following strategies as the holding period h increases. We report the annualised returns of the above strategies under different holding periods, for $h = \{1, 2, 3, 6, 9, 12, 18, 24, 30, 36, 48, 60\}$ months. From the figure, we can see that returns gradually decrease when moving from $h = 1$ to $h = 24$, indicating that the benefit of return continuation is largely offset by the subsequent time series reversal. The average returns remain stable after a holding period of more than 24 months, meaning that the reversal pattern stops after the end of the second year. The effect of the long-term reversal (36–60 month) uncovered in the previous section is weak, since it does not further reduce the mean returns. These results are consistent across the two strategies.

3.2. Timing of Time Series Reversal

If time series reversal ensues after the formation of trend-following signals, when does it begin and end? To answer this question, we implement TSM and RSM contrarian trend-following strategies using different time lags. This means that we take the opposite positions to those suggested by the trend-following strategy signals but using different lags from 2 to 36 months. Therefore, a total of 70 (35×2) lagged contrarian strategies are implemented using the trading signals with a holding period of $h = 1$.

Figure 4 shows a plot of the annualised mean returns of the aforementioned contrarian strategies, revealing the timing of the time series reversal. In the short term, contrarian strategies produce negative returns (2–10 months for TSM and 2–12 months for RSM). The negative short-term returns are followed by an intermediate reversal, which usually lasts for more than a year (12–26 months for TSM and 13–25 months for RSM). The portfolio returns become negative again after the intermediate-term reversal. None of these contrarian strategies produce statistically significant returns that outperform a naïve 1/N buy-and-hold strategy. Hence, simple contrarian trend-following strategies do not outperform the market.

Yao (2012) argues that January seasonality is one of the main reasons for contrar-